Discussion, April 14, 2016

What happens when you shine light on spinning cylindrical wire, about z-axis w/ angular velocity $\vec{\Omega} = \Omega \hat{z}$?

Recall, Maxwell's Eq^n for monochromatic light,

$$\vec{\nabla} \times \vec{E} = i\omega \vec{B} \qquad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = -i\omega \vec{D} \qquad \vec{\nabla} \cdot \vec{D} = 0$$

For linear media, $\vec{D} = \epsilon \vec{E}$ & $\vec{B} = \mu \vec{H}$
(at rest)

Above eqns needs to be modified for linear media moving w/ velocity v, $\frac{v}{c} \ll 1$, thus we will neglect $\left(\frac{v}{c}\right)^2$ or higher powers.

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 \vec{E} + (\varepsilon - \varepsilon_0)(\vec{E} + \vec{v} \times \vec{B})$$

$$\Rightarrow \quad \vec{D} = \varepsilon \vec{E} + (\varepsilon - \varepsilon_0)(\vec{v} \times \vec{B})$$

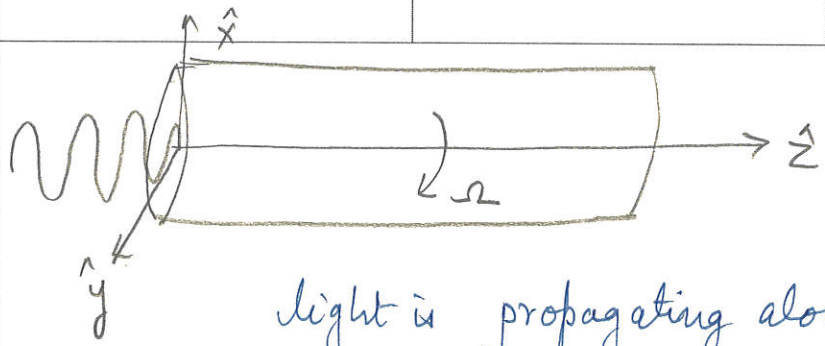
$$\text{or} \quad \underbrace{\vec{E} = \frac{\vec{D}}{\varepsilon} - \left(\frac{\varepsilon - \varepsilon_0}{\varepsilon}\right)(\vec{v} \times \vec{B})}_{} \qquad (1)$$

Magnetization, $\vec{M} = \vec{P} \times \vec{v} = (\varepsilon - \varepsilon_0) \vec{E} \times \vec{v}$

$$\Rightarrow \quad \vec{B} = \mu_0 \vec{H} + \mu_0 (\varepsilon - \varepsilon_0) \vec{E} \times \vec{v}$$

$$\Rightarrow \quad \underbrace{\vec{H} = \frac{\vec{B}}{\mu_0} - \frac{(\varepsilon - \varepsilon_0)}{\varepsilon} \vec{D} \times \vec{v}}_{} \qquad (2)$$

Now that we have expressions for $\vec{E}$ & $\vec{H}$ fields in terms of $\vec{D}$ & $\vec{B}$ fields, we can apply to solve our problem.



light is propagating along $\hat{z}$ w/ $\vec{E}$ & $\vec{B}$ fields in x-y plane.

$$\vec{v} = \vec{\Omega} \times \vec{r} \Rightarrow \vec{B} \times \vec{v} = \vec{B} \times (\vec{\Omega} \times \vec{r}) = \Omega(\vec{B}\cdot\vec{r}) - \cancel{\vec{r}(\vec{B}\cdot\vec{\Omega})}$$

$B \perp \hat{z}$ $\qquad \Rightarrow \vec{B} \times \vec{v} = \Omega(\vec{B}\cdot\vec{r})$

from ①, $\vec{\nabla} \times \vec{E} = \dfrac{\vec{\nabla} \times \vec{D}}{\varepsilon} + \left(\dfrac{\varepsilon - \varepsilon_0}{\varepsilon}\right) \underbrace{\vec{\nabla} \times (\vec{\Omega}(\vec{B}\cdot\vec{r}))}$

* First note that angular velocity, $\vec{\Omega}$ is a constant vector. So, $\vec{\nabla}$ explicitly operates on $\vec{B}(z)$ & $\vec{r}$.

* $$\left[\vec{\nabla} \times (\vec{\Omega}(\vec{B}\cdot\vec{r}))\right]_k = \epsilon_{ijk} \nabla_i (\Omega_j B_\mu r_\mu) = \Omega_j \epsilon_{ijk} [\nabla_i (B_\mu r_\mu)]$$

$$= \epsilon_{ijk} \Omega_j [(\nabla_i B_\mu) r_\mu + B_\mu \underbrace{(\nabla_i r_\mu)}_{\delta_{i\mu}}]$$

$$= \epsilon_{ijk} (\nabla_i B_\mu) \Omega_j r_\mu + B_i \epsilon_{ijk} \Omega_j$$

$$= (\vec{\nabla} B_\mu \times \vec{\Omega})_k r_\mu + (\vec{B} \times \vec{\Omega})_k$$

however, $\vec{\nabla} B_\mu = \dfrac{\partial B_\mu}{\partial z} \hat{z} \Rightarrow \dfrac{\partial B_\mu}{\partial z} (\hat{z} \times \vec{\Omega})_k r_\mu = 0$

bcoz $\vec{\Omega} = \Omega \hat{z}$

Thus,

$$\vec{\nabla} \times (\vec{\Omega}(\vec{B}\cdot\vec{r})) = \vec{B} \times \vec{\Omega}$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = \frac{\vec{\nabla} \times \vec{B}}{\epsilon} - \frac{(\varepsilon - \varepsilon_0)}{\varepsilon} (\vec{\Omega} \times \vec{B}) = i\omega \vec{B}$$ (from Maxwell's eqn)

Similarly, from ②, $\dfrac{\vec{\nabla} \times \vec{B}}{\mu_0} + \left(\dfrac{\varepsilon - \varepsilon_0}{\varepsilon}\right) \vec{\Omega} \times \vec{B} = -i\omega \vec{D}$

In fourier representation, $\vec{\nabla} \to i\vec{k}$, $\vec{k} = (0, 0, k)$

$(\vec{B}, \vec{D} \sim e^{ikz})$

$$\frac{i\vec{k}\times\vec{D}}{\varepsilon} - \frac{\varepsilon-\varepsilon_0}{\varepsilon}(\vec{\Omega}\times\vec{B}) = i\omega\vec{B}$$

$$\frac{i(\vec{k}\times\vec{B})}{\mu_0} + \frac{\varepsilon-\varepsilon_0}{\varepsilon}(\vec{\Omega}\times\vec{D}) = -i\omega\vec{D}$$

Multiply by '$-i$' both sides & take x & y components of both eqn, we get,

$$\left.\begin{aligned} &(1)\quad \omega B_x = -\frac{D_y k}{\varepsilon} - \frac{i(\varepsilon-\varepsilon_0)}{\varepsilon}\Omega B_y \\ &(2)\quad -\omega D_x = -\frac{B_y k}{\mu_0} + \frac{i(\varepsilon-\varepsilon_0)}{\varepsilon}\Omega D_y \end{aligned}\right\} \text{ x - component}$$

$$\left.\begin{aligned} &(1')\quad \omega B_y = \frac{k D_x}{\varepsilon} + \frac{i(\varepsilon-\varepsilon_0)}{\varepsilon}\Omega B_x \\ &(2')\quad -\omega D_y = \frac{k B_x}{\mu_0} - \frac{i(\varepsilon-\varepsilon_0)}{\varepsilon}\Omega D_x \end{aligned}\right\} \text{ y - component}$$

From the structure of eqns it is realized that we can define, $B_\pm = B_x \pm iB_y$

& $D_\pm = D_x \pm iD_y$

Now take (1) + i (1') & (2) + i (2') & combine fields,

$$\omega(B_x \pm iB_y) = \pm i\frac{k}{\varepsilon}(D_x \pm iD_y) \mp \frac{\varepsilon-\varepsilon_0}{\varepsilon}\Omega(B_x \pm iB_y)$$

$$\omega(D_x \pm iD_y) = \mp\frac{ik}{\mu_0}(B_x \pm iB_y) \mp \frac{\varepsilon-\varepsilon_0}{\varepsilon}\Omega(D_x \pm iD_y)$$

$$\Rightarrow \omega B_\pm = \pm\frac{ik}{\varepsilon}D_\pm \mp \frac{\varepsilon-\varepsilon_0}{\varepsilon}\Omega B_\pm$$

$$\omega D_\pm = \mp ik\epsilon_0 c^2 B_\pm \mp \frac{\varepsilon-\varepsilon_0}{\varepsilon}\Omega D_\pm$$

* Note that wave of given polarisation '+' do not mix with '−'.

$$\Rightarrow \quad B_\pm \left(\omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega \right) = \pm \frac{ik}{\varepsilon} D_\pm \qquad \text{—— (a)}$$

$$D_\pm \left(\omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega \right) = \mp ik \epsilon_0 c^2 B_\pm \qquad \text{—— (b)}$$

$$\Rightarrow \quad M \leftarrow \begin{pmatrix} \omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega & \frac{-ik}{\varepsilon} \\ ik\epsilon_0 c^2 & \omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega \end{pmatrix} \begin{pmatrix} B_\pm \\ D_\pm \end{pmatrix} = 0$$

Thus, for non-zero values of $B_\pm$ & $D_\pm$, $\det M = 0$ where M is a matrix denoted above.

$$\Rightarrow \quad \left(\omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega \right)^2 = \frac{k^2 \epsilon_0 c^2}{\varepsilon}$$

$$\text{or} \quad k_\pm = \frac{\sqrt{\varepsilon / \varepsilon_0}}{c} \left(\omega \pm \frac{\varepsilon - \varepsilon_0}{\varepsilon} \Omega \right)$$

Let $k_0 = \sqrt{\varepsilon/\varepsilon_0} \frac{\omega}{c}$ $\Rightarrow k_\pm = k_0 \pm \Delta k$, where

$$\Delta k = \frac{\varepsilon - \varepsilon_0}{c\sqrt{\varepsilon \varepsilon_0}} \Omega$$

Thus,
angle by which plane of polarisation of incident wave is rotated is given by, $\Delta\theta = \Delta k z$

After propagating distance L, $\Delta\theta = \frac{\varepsilon - \varepsilon_0}{c\sqrt{\varepsilon \varepsilon_0}} \Omega L$

Recall, $n = \sqrt{\varepsilon / \varepsilon_0}$, $\boxed{\Delta\theta = \frac{n^2 - 1}{n} \left(\frac{\Omega L}{c} \right)}$

Magnetic field, $\vec{B}(z) = A B_+ e^{ik_+ z} + B B_- e^{ik_- z}$

$B_+ = B_x \hat{x} + i B_y \hat{y}$ $\qquad$ A & B are constants.

$B_- = B_x \hat{x} - i B_y \hat{y}$

$$\Rightarrow \vec{B}(z) = A (B_x \hat{x} + i B_y \hat{y}) e^{i(k_0 + \Delta k) z} + B (B_x \hat{x} - i B_y \hat{y}) e^{i(k_0 - \Delta k) z}$$

Two natural modes of polarisation,

$B_+ = B_x\hat{x} + iB_y\hat{y}$, right circularly polarised

$B_- = B_x\hat{x} - iB_y\hat{y}$, left circularly polarised

Thus, we have two possibilities

$B_+ \neq 0, \quad B_- = 0 \Rightarrow B_x = iB_y$ or $B_y = -iB_x$

$B_- \neq 0, \quad B_+ = 0 \Rightarrow B_x = -iB_y$ or $B_y = iB_x$

$$\Rightarrow \vec{B}(z) = A(B_x\hat{x} + iB_y\hat{y})e^{i(k_0+\Delta k)z} + B(B_x\hat{x} - iB_y\hat{y})e^{i(k_0-\Delta k)z}$$

$$\Rightarrow \vec{B}(z) = AB_x(\hat{x} + i\hat{y})e^{i(k_0+\Delta k)z} + BB_x(\hat{x} - i\hat{y})e^{i(k_0-\Delta k)z}$$

Let at $z=0$, wave is linearly polarised in $\hat{x}$ dir^n,

$\Rightarrow \vec{B}(z=0) = \text{const}\ \hat{x} \Rightarrow B = A$

$$\Rightarrow \vec{B}(z) = \text{const.}\, e^{ik_0z}\left[(e^{i\Delta kz} + e^{-i\Delta kz})\hat{x} + i(e^{i\Delta kz} - e^{-i\Delta kz})\hat{y}\right]$$

$$\Rightarrow \boxed{\vec{B}(z) = 2\,\text{const.}\, e^{ik_0z}\left[\cos(\Delta kz)\,\hat{x} - \sin(\Delta kz)\,\hat{y}\right]}$$